Participant Profile Form

MSCA Staff Exchanges 2027 (MSCA-SE 2027) — consortium-building process

Unidad de Excelencia IOBA | University of Valladolid

Purpose:
This form is intended to collect structured information from interested participants for the preparation of an MSCA Staff Exchanges consortium. It is designed to support both (i) later completion of participant-related information requested in the EU Funding & Tenders Portal, and (ii) the creation of an internal profile database to help identify complementarities and facilitate contacts across the emerging consortium.

Please complete one form per:

- non-academic organisation (SMEs, NGOs , patient organisations)
- participating academic team (one form per PI, research-group, department).

This prospective **MSCA Staff Exchanges 2027 consortium** aims to build an international, interdisciplinary and inter-sectoral network in eye and vision research, centred on translational collaboration and staff mobility. Starting from Unidad de Excelencia IOBA's strengths in advanced therapies, biomarkers, inflammation, neuroprotection and health technology assessment, the network remains open to additional interdisciplinary collaborations, including areas such as ocular surface, retina, inherited and age-related eye disease, neuro-vision, optometry, drug delivery, AI-enabled diagnosis and patient-oriented validation, while remaining fully open to any other complementary expertise that could enrich the consortium and strengthen its patient-focused impact.

Important:
- please use this form to capture your general interests, expertise, and rough capacity to send and/or host staff.
- If you are not sure how to answer any field, you may leave it blank and highlight it.

1. Organisation and administrative profile

Organisation legal name	INM-Leibniz Institute for New Materials
Short name / acronym used in projects	INM
Participant Identification Code (PIC), if available	If no PIC is available yet, please indicate 'not yet available'.
Website(s)	www.inm-leibniz.inm.de https://www.leibniz-inm.de/en/research/scientific-units/dynamic-biomaterials/
Country	Germany
Organisation type	X Academic ☐ Non-academic ☐ Hospital / clinic ☐ SME ☐ Large company ☐ Research infrastructure ☐ NGO / foundation ☐ Public body ☐ Other
Main address	Campus D2 2; 66123 Saarbrücken, Germany
Department(s), institute(s), research group(s) or unit(s) that would be involved	Dynamic Biomaterials
Additional institutional resources relevant to collaboration	Core Facilities at INM for: • Electron Microscopy, https://www.leibniz-inm.de/en/research/core-facilities/electron-microscopy/ • Fluorescence Microscopy, https://www.leibniz-inm.de/en/research/core-facilities/fluorescence-microscopy/ • Accelerated Research Foundry, https://www.leibniz-inm.de/en/research/core-facilities/accelerated-research/ • Materials Physical Characterization • Bioprinting Laboratory

Main contact person for this profile	Aránzazu del Campo, Prof. Scientific Director and Department Head Dynamic Biomaterials
Main contact email	Aranzazu.delcampo@leibniz-inm.de
Other contact person(s), if relevant	Martina Bonnard, Secretary, martina.bonnard@leibniz-inm.de

2. Scientific / innovation profile and relevance for an IOBA-led consortium

Please describe your profile at the level of the participating team / unit most relevant to the prospective consortium.

Indicative thematic relevance (tick all that apply)	
☐ Ocular surface	☐ Retina
☐ Inherited retinal dystrophies (IRDs)	☐ Age-related ocular disease
☐ Neuro-vision / integrated vision / neuro-ophthalmology	☐ Optometry / visual behaviour / rehabilitation
X Advanced therapies / regenerative medicine	☐ Gene therapy / cell therapy / tissue engineering
☐ Inflammation / biomarkers / neuroprotection	X Drug delivery / formulations / nanomedicine
☐ Drug discovery / screening / repurposing	☐ Ocular vascular disease
☐ Infection / trauma-related ocular disease	☐ Ocular pain / quality of life / patient-reported outcomes
☐ Health technology assessment	☐ AI-assisted diagnosis / software / digital tools
☐ Clinical trials / translational validation	☐ Other (please specify):

Free Keywords for internal searchable database (please provide up to 10)	Ocular devices, sustained drug delivery, contact lenses, dry eye, biomaterials,
Geographies of interest for exchanges (optional)	

Overview of your research / innovation focus
The Research Department *Dynamic Biomaterials* at INM develops bioactive materials to encapsulate and instruct living cells. With our competences in hydrogel synthesis, photoswitches, biofabrication technologies, and cell biology, we create synthetic cellular microenvironments with dynamic properties and study how living and non-living matter interact. We exploit such interactions to regulate cellular functions by materials design. In cooperation with synthetic biologists and clinicians around the world, we contribute to the field of wearable/implantable living therapeutic devices for sustained and sustainable treatment of chronic diseases, both from fundamental research and application perspectives. These devices contain material-encapsulated drug biofactories that support on-site (in the body) production of the drug using energy sources from the environment. They represent a new concept for zero-waste pharmacology. We work on (i) understanding how material design affects bioactivity of biofactories, (ii) the parallelization of in vitro testing of living devices in simulated physiological conditions to predict bioactivity variability and changes in disease conditions, and on (iii) the design of advanced prototypes for technology transfer, with living drug-eluting contact lenses as advanced prototypes towards innovation.

Why your team could be relevant to a translational eye-health and vision consortium
- Expertise in biomaterials for application in ophthalmic devices

- Broad experience in cell-materials interactions and their impact in biocompatibility
- Experience in the design (from TRL 0) of biomaterials-based ophthalmic devices for the ocular surface
- Networked with contact lens industry and ocular pharma
- Currently going up the technology transfer ladder with an innovative drug eluting contact lens. Experience and network with innovators in ocular market
- Interested in networking with ophthalmic practitioners and partners in the ophthalmic market

3. Expertise, assets and collaboration interests

Core expertise, methods and platforms you could contribute

- Expertise in biomaterials synthesis and purification: organic and hydrogel chemistry
- Expertise in hydrogel processing: soft-lithography, biofabrication (extrusion printing, volumetric printing, laser-assisted printing, ink-jetting)
- Expertise in soft materials characterization: mechanical, thermal, diffusion and optical properties
- Network of innovation experts and innovators in ocular devices

Infrastructure, equipment, facilities or datasets relevant to consortium work

- **Biofabrication equipment**
  - **Extrusion bioprinting:** CELLINK BIO X, GeSiM Bioscaffolder, RegenHU 3D Discovery, customized multimaterial thread fabrication station
  - **Electrospinning:** Bioinica Fluidnatek LE50
  - **Microfluidics:** Atrandi ONYX, Fluigent Raydrop
  - **Film coating and disposition:** Ossila Slot Die Coater
  - **Automated liquid handling:** Hamilton Microlab Nimbus
  - **Volumetric Printing:** Readily 3D Tomolite Performance 2.0
  - **Two-Photon lithography:** Nanoscribe Photonic Professional GT2
- **Optical property characterization equipment**
  - **Optical coherence tomography (OCT):** Thorlabs Ganymed OCT
  - **Refractive index measurement:** Anton Paar Abbemat Multiwavelength Refractometer
- **Mechanical characterization equipment**
  - **Rheological characterization:** Anton Paar MCR 702e, TA Instruments DHR3 Rheometer
  - **Nano-mechanical testing:** Optics11 Pavone Nanoindenter
  - **Dynamic mechanical analysis (DMA):** TA Discovery DMA 850
- **Fluorescence microscopy equipment**
  - **Total Internal Reflection (TIRF) microscopy:** Nikon TiE-TIRF
  - **Automated live-cell imaging and high-content microscopy:** ZEISS Celldiscoverer 7
  - **Light-sheet fluorescence microscopy:** ZEISS Lightsheet Z.1
  - **Super-resolution and nonlinear optical microscopy:** ZEISS LSM 880 Airyscan NLO
  - **Live-cell imaging and photomanipulation:** ZEISS Observer Z1 with Rapp OptoElectronic system
- **Electron microscopy equipment**
  - **Environmental scanning electron microscopy (ESEM):** FEI Quanta 250 FEG, FEI Quanta 400 FEG
  - **Focused ion beam-scanning electron microscopy (FIB-SEM):** FEI Versa 3D
  - **Transmission electron microscopy (TEM):** JEOL JEM-2100
- **Accelerated research facility**
  - **High-throughput liquid handling:** Echo 650 Acoustic Droplet Ejection System, Hamilton STAR
- **Electronic Documentation**
  - Aligned Elements
  - RSpace

Training capacity and knowledge-transfer potential

INM has a dedicated facility with specialized infrastructure for automating living therapeutic materials prototype fabrication and upscaling, including highly qualified personnel to run the interdisciplinary facilities and to support the establishment of a quality management system.

What your team would hope to gain from participating
- *Ophthalmology practitioners to provide feedback and eventually test our prototypes* - *In vitro and in vivo models of ocular tissues and pathological states* - *Experience in patients surveys, contact to patients and patient associations* - *Industrial insight* - *Regulatory knowhow* - *Long term strategic collaboration partners in the field*

4. Key people likely to be involved

Please list the key people who may be relevant to the consortium at this stage. *This is not a final, binding or secondment list.*

Name	Position / role	Department / unit	Staff category	Career stage	Email / identifier (optional)

Main disciplines represented in your team	Biomaterials
Experience in EU / international / multi-partner collaborative projects	Participant of FET-OPEN and ITN Projects
Experience with clinical, translational or patient-facing research (if relevant)	☐ Yes X No ☐ Not applicable | Short note:

5. Evidence of capacity and track record

Up to 5 relevant publications / datasets / software / products / services / other achievements
Z. Kafrashian, J. Feng, and A. del Campo, *Fabrication Technologies for Soft, Multimaterial Optical Fibers for In Vivo Diagnostics and Phototherapy, With a Focus on Extrusion Printing.* *Advanced Materials Technologies 2026, 11, 9, e01789* *https://doi.org/10.1002/admt.202501789* *K. Desai, S. Sankaran, A. del Campo, S. Trujillo* *A screening setup to streamline in vitro engineered living material cultures with the host* *Materials Today Bio 2025, 30, 101437* *https://doi.org/10.1016/j.mtbio.2024.101437* *M. Puertas-Bartolomé, I. Gutiérrez-Urrutia, L. L. Teruel-Enrico, C. N. Duong, K. Desai, S. Trujillo, C. Wittmann, A. del Campo,* *Self-Lubricating, Living Contact Lenses.* *Advanced Materials 2024, 36, 2313848* *https://doi.org/10.1002/adma.202313848* *A. Rodrigo-Navarro, S. Sankaran, M.J. Dalby, A. del Campo, M. Salmeron-Sanchez,* *Engineered Living Biomaterials* *Nat Rev Mater 2021, 6, 1175* *https://doi.org/10.1038/s41578-021-00350-8* *S.Bhusari, S.Sankaran, A.delCampo* *Regulating Bacterial Behavior within Hydrogels of Tunable Viscoelasticity.* *Adv. Sci. 2022, 9, 2106026* *https://doi.org/10.1002/advs.202106026*

Up to 5 relevant previous projects, grants, networks or collaborations
DFG-SPP2451, Living Materials with Adaptive Functions (2025-28), PI and Coordinator *Leibniz SAW-Transfer, Transfer Readiness of a New Technology for Sustained Ocular Drug Delivery, PI and Coordinator* *BMFTR-MaterialVital Hub, MaterialVital Digital, PI* *Leibniz Science Campus Living Therapeutic Materials (2020-28), PI and Coordinator*

Patents, protected know-how, clinical protocols, validated tools or other translational assets (if any)
Selbst-schmierende Kontaktlinsen. Filed on 09.11.2023 **(102023131356.8)** Lebendes Kompositmaterial für medizinische Produkte / Biokompatibles Polymer. Filed on 09.11.2023 **(102023131358.4)** Neuartige Hydrogele / Hydrogele Thio-methylsulfonyl Netzwerke für Biomaterialien, **EP 3994202 A1, JP 2022539551 A, KR 1020220031906 A, US 20220235316 A1** Leitfähige funktionalisierbare Nanokomposite / Leitfähige Tinten zur Herstellung von Elektroden, die spezifisch mit bioaktiven Molekülen, Zellen oder Geweben wechselwirken. **20129135645 A1, JP 2022539551 A, KR 1020220031906 A, US 20220235316 A1**

6. Staff-exchange capacity (only rough estimation at this stage)

Please provide only a rough, non-binding estimate of your organisation's/team's capacity to send and/or host staff **over the 48-month duration of the project.**

Could your organisation/team second staff?	X Yes ☐ Maybe ☐ No capacity to send staff on secondment
Could your organisation second staff categories beyond researchers (where relevant to R&I activities)?	X Only Researchers X Technical Staff/Innovation ☐ Managerial / project support ☐ Admin Staff ☐ Other profiles
Could your organisation host incoming staff from academic and/or non-academic sectors?	X Academic only ☐ Non-academic only ☐ Both ☐ No capacity to host
Administrative / HR / legal considerations that may affect mobility	

7. Matchmaking information: Summary for the internal participant database

Capabilities you would most like other interested participants to notice
- Biofabrication - Contact Lenses

Capabilities or expertise you would especially like to access through the consortium
- Clinical workflows - In vitro and in vivo models - Patient surveys - Manufacturing of ocular devices

Please Confirm:

Open to being contacted by other interested participants through the coordination team's mediation?	X Yes ☐ No

☐ We understand that the information contained in this form will be searchable and accessible to other interested participants within the pre-consortium process.

8. Final remarks and query items

Please note any uncertainties, prerequisites, internal approvals needed, or items that should be considered further before the consortium is consolidated.

Internal use and confidentiality statement

The information provided in this form will be used only for internal consortium-building purposes in connection with the preparation of a prospective MSCA Staff Exchanges proposal (2027 call) coordinated from Unidad de Excelencia IOBA / University of Valladolid. It may be used by the coordination team to: (i) assess strategic fit and administrative readiness, (ii) prepare participant-related information required at proposal stage, and (iii) build an internal profile database to facilitate contacts among interested participants. The information will not be shared outside this pre-consortium context without prior permission from the contributing organisation / contact person, except where later required for the formal preparation of a proposal with the explicit involvement of that organisation. Submission of this form confirms that the contributor agrees to this internal use. Organisations may request correction, update or withdrawal of their information at any time by contacting the coordination team.

X Understood

Name: Aránzazu del Campo
Date completed: 27.08.2025

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